

Thermal desorption spectroscopy (TDS)

In this experiment, we will determine the binding energy of a molecule to a surface or, in a thicker film, its sublimation enthalpy by means of thermal desorption spectroscopy (TDS), also called TPD (temperature programmed desorption). This experiment is rather simple, but allows a quantitative determination of the surface binding energy of an adsorbate, which is an important parameter for describing the bonding of the atom or molecule to the surface.

Please bring a USB drive (smaller than 32 GB) and a notebook that runs a data analysis software (e.g., SciDAVis or Origin) so that you are able to check your spectra during the experiment.

Effective: 6/19/2024

1. General preparation

Read up on the following terms in books and/or online:

partial pressure measurement, pumping speed, construction and working principle of a quadrupole mass spectrometer (e.g. [5]), adhesion factor, dose, Polanyi-Wigner equation, desorption of 0th, 1st, and 2nd order, evaluation methods for thermal desorption spectra.

2. Theoretical background

2.1. Theoretical aspects

Thermal desorption spectroscopy (TDS) or temperature programmed desorption (TPD) measures kinetic rates for desorption processes. First, a specific dose of a gas is applied. By heating the sample (usually with a constant heating rate, which leads to the name "temperature programmed") the adsorbed particles are desorbed from the surface. Typical energies for desorption are between 10 kJ mol⁻¹ and 200 kJ⁻¹ mol. The desorption process is monitored via a mass spectrometer measuring the increasing corresponding partial pressure and is recorded as a function of the sample temperature or time.

If the pumping speed of the vacuum system is sufficiently high, the partial pressure p_{partial} of the desorbing species is proportional to the desorption rate:

$$r_{\text{des}} = -\frac{d\theta}{dt} \sim p_{\text{partial}}$$

R_{des} . For the details we refer to the script of the lecture.

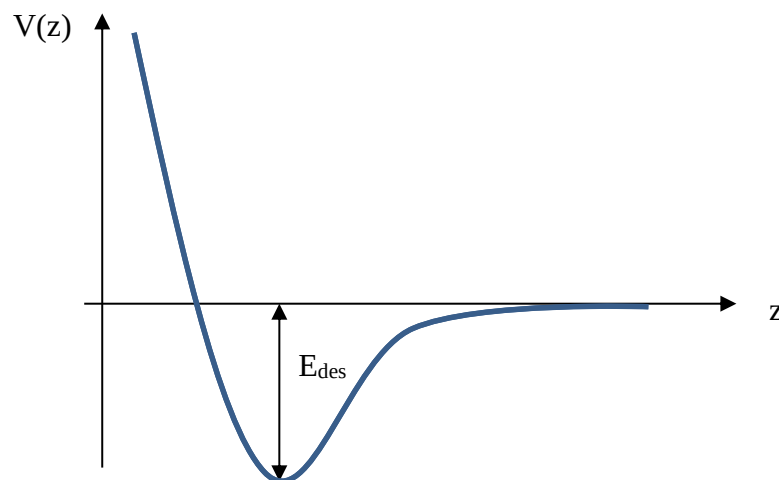
Thus, the areas under the pressure-versus-time or pressure-versus-temperature curves (TPD curves) are proportional to the number of desorbed particles. Usually it is assumed that the partial pressure is sufficiently small and re-adsorption is therefore excluded.

Very commonly TPD spectra are evaluated by using the *Polanyi-Wigner equation*:

$$r_{\text{des}} = \nu_0 \cdot \theta^m \exp(-E_{\text{des}} / k_{\text{B}}T). \quad (1)$$

From the pressure/temperature curves, the desorption energy E_{des} , the pre-exponential factor ν_0 of the Polanyi-Wigner equation and the desorption order m (0, 1, or 2) of the desorption process can be determined. Please note that in general the dimension of ν_0 depends on the dimension of θ and also m . (Note that ν_0 is named k_0 in the lecture script.)

Furthermore, it is sometimes possible to assign several desorption processes (marked by different desorption maxima) to different adsorption states of an adsorbate. In the simplest case these would be different sites. The *Polanyi-Wigner equation* is actually a microscopic equation describing a rate. It can partially be derived from the transition state theory. The desorption energy is the energy required to desorb the particle from bonded state on the surface. It is indicated in the schematic figure below.



Sometimes the desorption energy is also named as *adsorption enthalpy*, because it is related to the heat that is released during the adsorption process of a particle. This is not quite correct, since E_{des} is a microscopic quantity describing a potential and is obtained from a kinetic experiment, like TPD, whereas the adsorption enthalpy ΔH_{ad} is a quantity related to an ensemble of particles and used to describe thermodynamic equilibrium. For instance, it can be determined by applying the Clausius-Clapeyron equation to the situation where the stationary gas pressure above an adsorbed film is measured at constant temperature. For a non-activated adsorption (see literature), the difference

between E_{des} and ΔH_{ad} is, however, not very large, namely of the order of $R T$, which is approximately 2.5 kJmol^{-1} . This difference can be neglected in many cases. The same holds true when the sublimation enthalpy is determined from TPD experiments desorbing multilayers of particles from the sample.

2. Experimental setup

2.2. *Experimental setup*

2.2.1. The UHV-system

An ultra-high vacuum system is available for carrying out the experiment (see Figure 1). The system contains a turbo molecular pump (TP) with a nominal pumping speed of 300 l/s. After several hours of baking, a pressure lower than 1×10^{-9} mbar can be achieved. For pressure measurement in the UHV range an ionization gauge (not shown in Fig. 1) is used.

The dosing of gas takes place via leak valves, which allow an adjustment of defined leak rates. The required chemicals (phenol, benzene) are held in evacuated glass flasks. At room temperature they have sufficient vapor pressure to be leaked into the vacuum chamber without further heating.

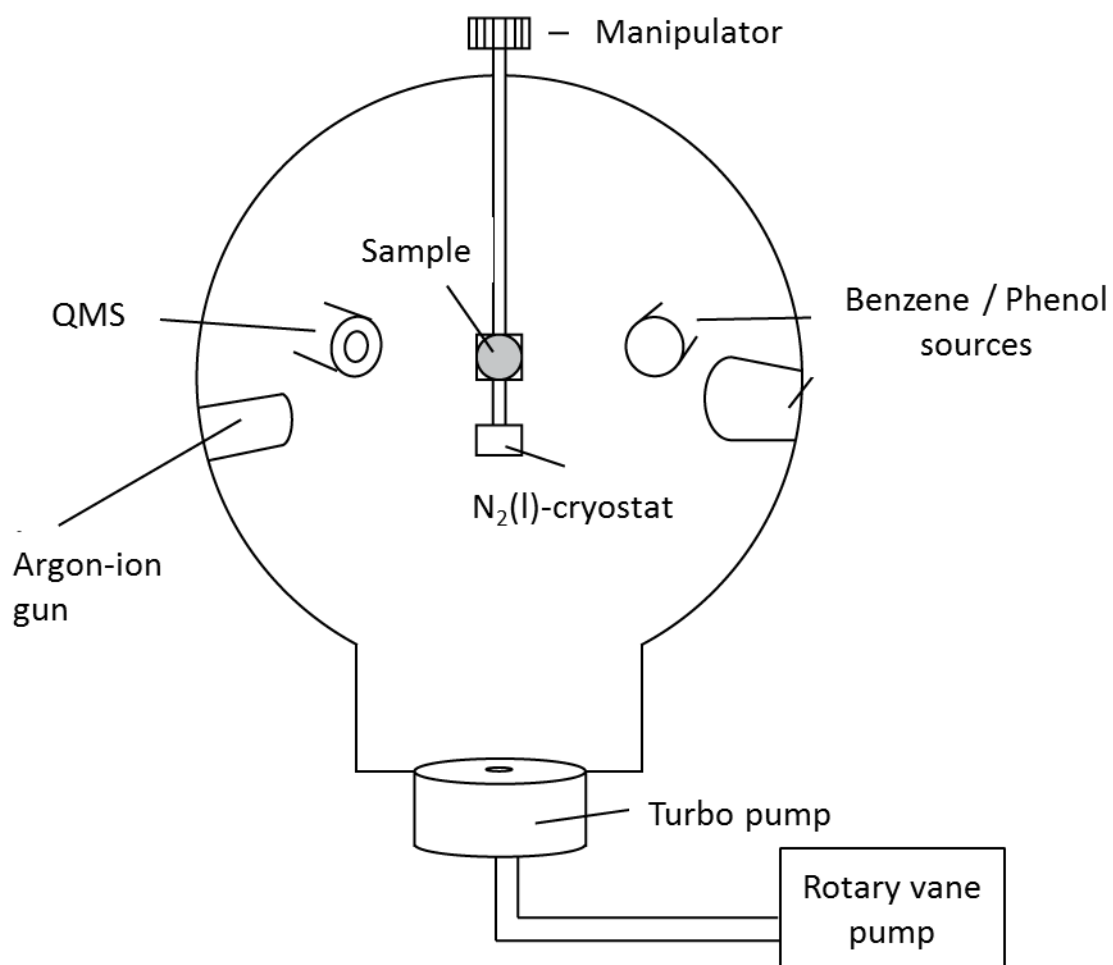


Figure 1: Schematic setup of the UHV-system.

2.2.2. Measuring equipment

The metal sample (here an Ag(111) crystal) is attached to a moveable manipulator in the center of the UHV chamber (see Fig. 1). The following devices are mounted on the lateral flanges:

- an **argon-ion gun** for sample cleaning. Figure 3 show the actual setup and a schematic illustration. The energy of the incident Ar^+ ions can be set between 100 eV and 3000 eV and the ion current between 0.1 μA and 10 μA .
- a **mass spectrometer (QMS)** for residual gas analysis and for thermal desorption spectroscopy. The mass spectrometer uses a quadrupole mass filter, whose basic structure is shown in Figure 4.

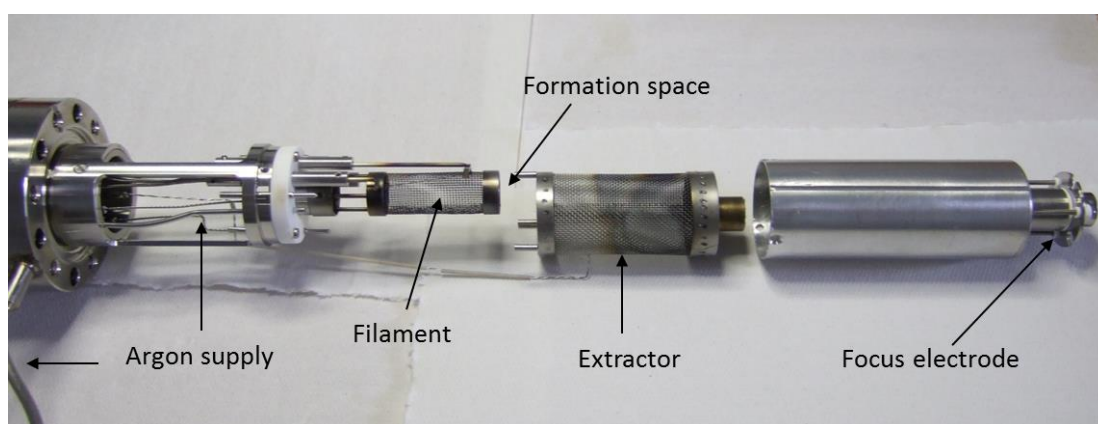


Figure 2: Setup of the argon-ion gun.

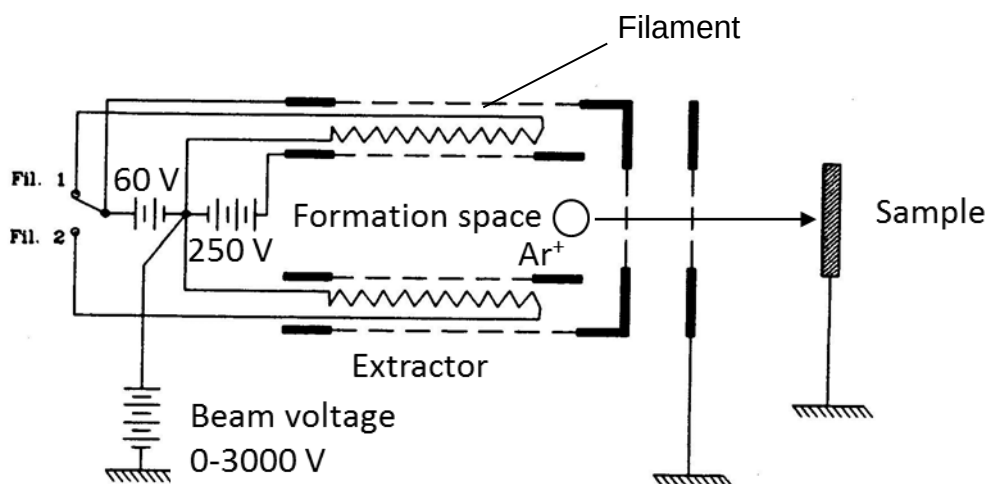


Figure 3: Schematic drawing of the argon-ion gun.

The mass filter consists of four cylindrical rods arranged in a square. Two opposite bars are electrically connected (see Fig. 4). Between the bars a voltage is applied which contains a DC (U) and an AC voltage component ($V\cos(\omega t)$), resulting in $V(t) = U + V \cdot \cos(\omega t)$.

In the QMS ion source, molecules are ionized by electron bombardment. The formed cations then enter the rod system. Considering first a rod system without superimposed AC component, the cations are repelled by the positively charged rods and attracted

by the negative ones. Since no other forces act on the ions, all ions get neutralized on the negative rods. No ions pass the rod system. Now, if the DC voltage is superimposed with an AC voltage, the ions start oscillating perpendicular to the axis of the filter. Ions with a certain oscillation amplitude will now be able to pass the rod system: their oscillation amplitude is small enough to not hit the negative rods and get neutralized, nor are they deflected so much that they hit the positively charged rods. The oscillation amplitude, i.e. the deflection of the ions, depends on the frequency and the amplitude of the AC voltage, as well as the mass and charge of the ion. The equations of motion of the ions near the axis of the filter result in an image of the entrance aperture onto the exit aperture for $\frac{m}{e} = \text{const.} \frac{V}{\omega^2}$. This means at a constant frequency it is possible to allow a passing of different masses (i.e. different ions) through the rod system by varying the AC amplitude V . Since the range of m/e of the transmitted region depends in addition on the ratio $\frac{U}{V}$, this is kept constant during a mass scan.

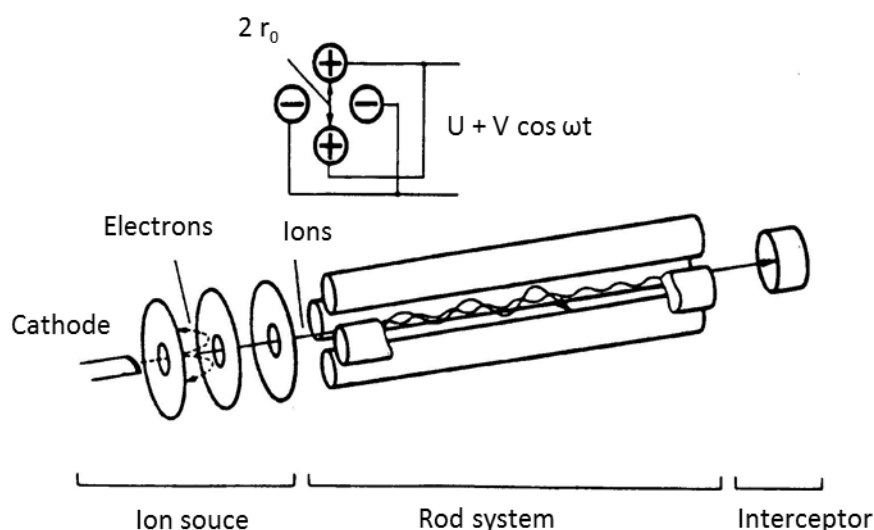


Figure 4: Schematic drawing of the quadrupole mass filter.

2.2.3. Sample holder

The Ag(111) crystal is attached to a manipulator with which the sample can be rotated about 260° . The positions for all measurements or treatments are thus reproducible. The sample is mounted on a molybdenum alloy crystal holder. The crystal can be heated by means of a tungsten filament which is located inside the crystal holder. To cool the sample, a copper strand is attached to the backside of the holder. The copper strand is connected to a liquid nitrogen tank. This allows the sample to be cooled to a temperature of <90 K. The entire crystal holder is electrically isolated from the vacuum chamber. The insulation is achieved by ceramic insulators. Since normal ceramics

usually conduct heat poorly, the copper strand is isolated from the liquid nitrogen container using a sapphire disc. Sapphire has very good and increasing thermal conductivity at low temperatures. This setup achieves very high heating and cooling rates. The temperature can be measured by a NiCr-Ni thermocouple (also known as type K) attached to the edge of the sample. The evaluation of the thermoelectric voltage is done by a control unit (PID temperature controller, "Eurotherm"). The Eurotherm also regulates a DC power supply. Depending on the settings of the Eurotherm, this allows TDS to be performed at a constant rate of heating and to maintain a set temperature (e.g. for sample preparation).

2.2.4. Sample preparation

The sample was polished with diamond abrasive paste of decreasing grain size (down to 1 μm). The prepared sample is integrated into the UHV system. By argon ion bombardment (ion current about 1 - 10 μA , accelerating voltage 800 V, for 20 min) and subsequent annealing under vacuum ($p < 1 \times 10^{-7}$ mbar) at 600 °C for 45 min chemical contaminations of the surface are eliminated and mechanical damages to the surface are healed.

2.2.5. Electronics rack

The electronics required for the experiment are built into a 19" rack, with the exception of the computer (see Figure 5). In addition to the mass spectrometer and the argon ion gun controls, the temperature controller ("Eurotherm") as well as the pressure gauges are located in this rack. Furthermore there is an interlock-system in the rack which, in the event of a power failure or turbo pump failure, puts the UHV system in a safe operating condition by closing a large gate valve.

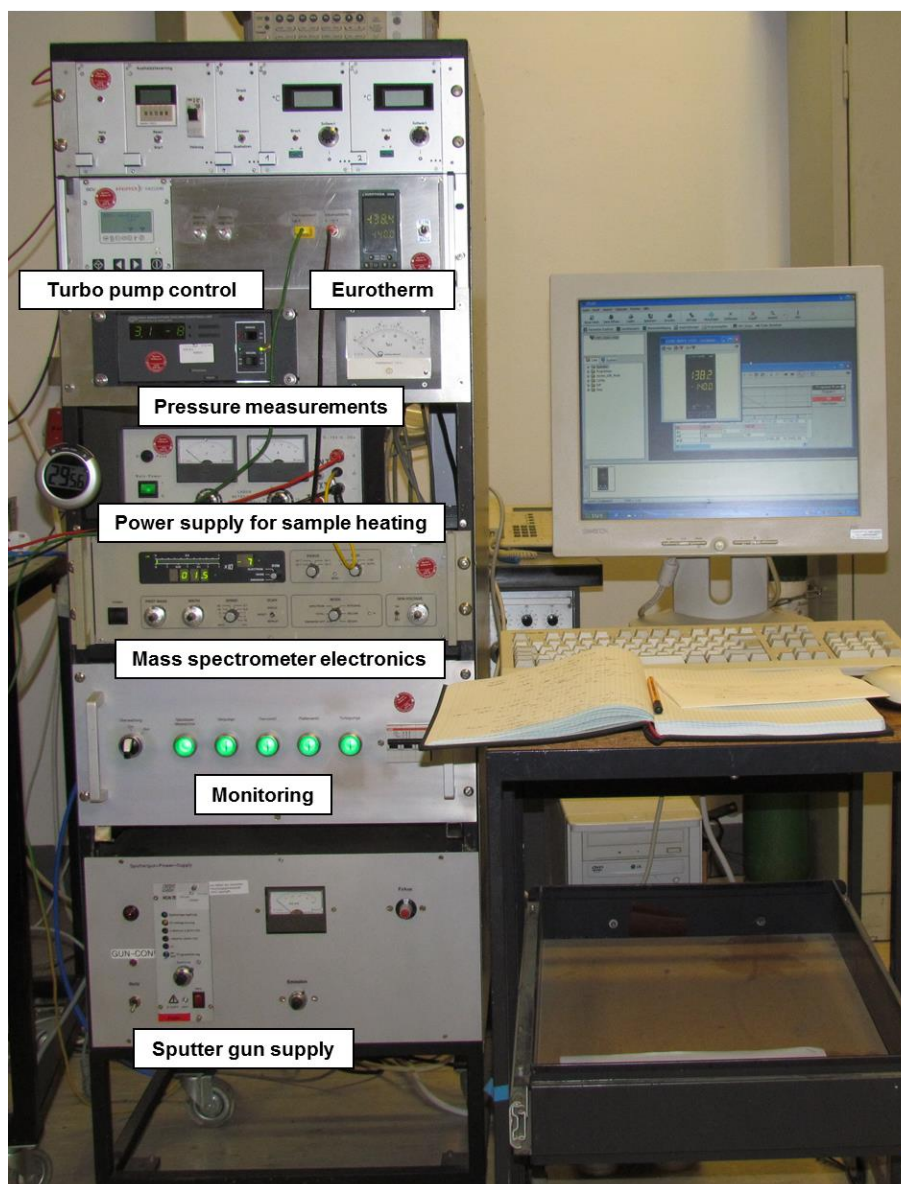


Figure 5: Control electronics and measuring computer.

3. Experimental Procedure

3.1. *Residual gas analysis*

Measure the current pressure using the ionization manometer.

Make yourself familiar with the dosing valves. Dose argon into the chamber until the pressure has risen to about 1×10^{-8} mbar. Close the valve. Watch the pumping behavior of the chamber.

Together with your supervisor, put the quadrupole mass spectrometer (QMS) into operation. Record a residual gas spectrum. Assign all the peaks.

Briefly degas the filament of the ionization manometer and record a mass spectrum at the same time. What is changing?

3.2. *Mass spectra*

Record the mass spectra of phenol and benzene (p at about 1×10^{-8} mbar). Index all peaks. Identify the main peak. Watch the pumping behavior of the chamber after closing the valves. Is there a difference in comparison to argon?

3.3. *Sample preparation*

Sputter the Ag(111) sample. Then anneal the sample. Cool down the sample afterwards. Use well-proven parameters from the lab book for each step. Regularly check the nitrogen, it must not run out during preparation and experiment.

3.4. *Recording of TDS spectra*

You should record as many TDS spectra as possible with different initial conditions of the sample. Observe whether the conditions regarding pressure and heating rate are met.

Together with your supervisor, deposit phenol/benzene to the sample. Measure the TDS spectra.

On the basis of your already recorded TDP spectra (check with the data analysis program on your notebook) decide which further doses make sense. The aim of the experiment is not only to determine the desorption energies of monolayer and multilayer, but also to calibrate the optimal dose to achieve a closed monolayer.

4. Evaluation

4.1. *Residual gas analysis*

1. Index all peaks in the spectra of the residual gas, benzene and phenol.

4.2. *TPD spectra*

1. Plot all TPD spectra into one single diagram. Describe the spectra and their dependency on the dosage. What are the origins of the different peaks?
2. Try to identify the spectrum that corresponds as best as possible to a complete monolayer of molecules. Estimate the number of molecules in the monolayer (in cm^{-2}) on the basis of a geometric structure model assuming that the molecules lie flat on the surface.
3. Determine the area under all TPD spectra and calibrate them to determine the absolute coverages, i.e. the number of molecules per cm^2 . These are the start coverages (θ_0).
4. Plot the obtained start coverages θ_0 versus the applied dosages (in mbars). Do you find a linear behavior?
5. Estimate the collision rate Z for the molecule you adsorbed (see [1] eq. (1.52)) and convert the dosages into the number of molecules that have impinged onto the surface (in cm^{-2}). Try to give an estimate on the sticking factor s in the limit of small coverages.

4.3. *Evaluating the multilayer desorption*

1. Analyze the multilayer TPD spectra to determine the activation energy for the desorption of the multilayer. For this purpose, use the logarithmic form of the Polanyi-Wigner equation for 0th order ($m = 0$). Plot $\ln(r)$ versus $1/T$. This plot is usually referred to as an Arrhenius plot. However, this plot is somewhat confusing, because contrary to the experiment, the high temperatures are found at the left end of the plot. Thus, to make the plot more intuitive, plot the $1/T$ axis from right to left. Determine the desorption energy E_{des} and the pre-exponential factor ν_0 . For the determination of the pre-exponential factor ν_0 you need to calibrate the desorption rate in number of monolayers molecules per second.
2. Compare your results with those reported in the literature ([3, 4]).

4.4. Evaluating the monolayer desorption

1. Plot all spectra for start coverages up to one monolayer into one diagram.
2. We expect the monolayer desorption to be of first order ($m = 1$). One signature of a first order desorption peak is that the peak temperature (T_p), i.e. the temperature where the desorption rate is maximal, is independent from the start coverage θ_0 (see [1] eq. (2.44)). (This is not the case for zero and second order desorption.) Please check whether this is fulfilled here.
3. The determination of the desorption energy from a first order desorption spectrum is not as straightforward as for the multilayers. The reason is that the coverage enters into eq. (1) as an additional parameter. As a first evaluation, you can determine E_{des} from the position of the peak temperature T_p according to the formula that was given by Redhead [6] (cf. also [1] eq. (2.45)):

$$E_{\text{des}} = k_B \cdot T_p [\ln(\nu_0 T_p \beta^{-1}) - 3.64]. \quad (2)$$

(Note the typo in Attard! It should be 3.64 not 3.46). ν_0 denotes the frequency factor and β the heating rate in K per s (cf. eq. (1)). Eq. (2) is valid for a certain range of the parameter $10^{13} > \nu_0 \beta^{-1} > 10^8 \text{ 1/K}$. In order to use the formula we assume $\nu_0 = 10^{13} \text{ s}^{-1}$.

4. A second possibility to evaluate the data is to simulate the desorption curves. For this purpose you can use one of the script that are provided on the eVampus page and which run under octave, python or in the simplest case Wolfram alpha.

Please adjust the parameters ν_0 and E_{des} such that the script calculates the desorption rates (in ML per s) for three different initial coverages between 0 and 1 ML.

5. Compare your results with those given in the literature ([3, 4]).

Literature

- [1] G. Attard, C. Barnes, *Surfaces*, Oxford Science Publications, Oxf. Univ. Press, 1998, page 71.
- [2] W. Kolasinski, *Surface Science*, Wiley and Sons, New York, page 217 ff.
- [3] J. Lee, S. Ryu, S.K. Kim, *Surface Science* **481** (2001) page 163 - 171. (phenol)
- [4] T.J. Rockey, M. Yang, H.-L. Dai, *Journal of Physical Chemistry B* **110** (2006) page 19973–19978 (benzene)
- [5] http://www.chemgapedia.de/vsengine/vlu/vsc/de/ch/3/anc/masse/ms_massenanalysator_quadrap.vlu.html
- [6] P.A. Redhead, *Vacuum* **12** (1962) page 203.